

Homework 2.2

Problem 3

How many digits of precision are lost in the subtraction $1 - \cos(x)$ when $x = \frac{1}{4}$?

Using the Loss of Precision theorem, we get that:

$$1 - \frac{y}{x} \geq 2^{-p}$$

Where p is the number of digits lost. Let $y = \cos(x)$ and $x = 1$ then:

$$1 - \frac{\cos(x)}{1} \geq 2^{-p}$$

$$1 - \cos\left(\frac{1}{4}\right) \geq \frac{1}{2^p}$$

[Did not finish]

Problem 4

(Continuation) For the function in the preceding problem, find a suitable Taylor Series by which it can be accurately computed.

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2n!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$$

$$1 - \cos(x) = \frac{x^2}{2!} - \frac{x^4}{4!} + \dots$$

Using the approximation $1 - \cos(x) \approx \frac{x^2}{2} - \frac{x^4}{24}$, we would have that this would be a great approximation for x around 0 since x^6 and higher order terms would be almost 0.

Problem 9

Suggest ways to avoid loss of significance in these calculations.

a) $\sqrt{x^2 + 1} - x$

As $x \rightarrow \infty$ we have that $f(x) = \sqrt{x^2 + 1}$ and $g(x) = x$ are almost the same value so for large values of x , it would be wise to do the following:

$$\left(\sqrt{x^2 + 1} - x\right) * \left(\frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1} + x}\right) = \frac{x^2 + 1 - x^2}{\sqrt{x^2 + 1} + x} = \frac{1}{\sqrt{x^2 + 1} + x}$$

This avoids having to subtract to close numbers.

b) $\log(x) - \log(y)$

For $x \approx y$, we would be subtracting by nearly equal numbers, thus to avoid this:

$$\log(x) - \log(y) = \log\left(\frac{x}{y}\right)$$

Since division works nicer for us, dividing the numbers and then taking the log will give us more accurate numbers.

c) $x^{-3}(\sin(x) - x)$

For small values of x , we know that $\sin(x) \approx x$ so in order to avoid this issue, we can do the following:

$$\begin{aligned}\sin(x) &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \\ \Rightarrow \sin(x) - x &= -\frac{x^3}{3!} + \frac{x^5}{5!} + \dots \\ \Rightarrow \frac{(\sin(x) - x)}{x^3} &= -\frac{1}{3!} + \frac{x^2}{5!} + \dots = \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-2}}{(2n+1)!}\end{aligned}$$

Thus around $x = 0$, we would use this Taylor series and truncate it at whichever point needed for whatever accuracy we desire.

d) $\sqrt{x+2} - \sqrt{x}$

For large values of x , we would be subtracting almost identical numbers, thus we do the following:

$$(\sqrt{x+2} - \sqrt{x}) * \left(\frac{\sqrt{x+2} + \sqrt{x}}{\sqrt{x+2} + \sqrt{x}} \right) = \frac{x+2-x}{\sqrt{x+2} + \sqrt{x}} = \frac{2}{\sqrt{x+2} + \sqrt{x}}$$

e) $e^x - e$

When $x \rightarrow 1$, these numbers will be nearly identical, so to avoid subtracting by like numbers:

$$(e^x - e) * \frac{e^x + e}{e^x + e} = \frac{e^{2x} - e^2}{e^x + e}$$

This works since e^2 is now farther away from e^{2x} when $x \rightarrow 1$ and adding positive numbers in the denominator causes little issues.

f) $\log(x) - 1$

When $\log(x) \rightarrow 1 \Rightarrow x \rightarrow e$ we would be subtracting similar numbers, to avoid this:

$$\log(x) - 1 = \log(x) - \log(e) = \log\left(\frac{x}{e}\right)$$

g) $\frac{\cos(x) - e^{-x}}{\sin(x)}$

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{4!} + \dots$$

$$e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{3!} + \frac{x^4}{4!}$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

$$\Rightarrow \cos(x) - e^{-x} = x + \frac{x^3}{3!} + \frac{x^5}{5!}$$

For small x , we have that $x^n \approx 0, n \geq 4$, thus

$$\frac{\cos(x) - e^{-x}}{\sin(x)} \approx \frac{x + \frac{x^3}{3!}}{x - \frac{x^3}{3!}}$$

h) $\sin(x) - \tan(x)$

$$\begin{aligned} \sin(x) - \tan(x) &= \sin(x) - \frac{\sin(x)}{\cos(x)} = \sin(x) \left[1 - \frac{1}{\cos(x)} \right] = \sin(x) \left[\frac{1 - \cos(x)}{\cos(x)} \right] \\ &= \sin(x) \left[\frac{1 - \cos^2(x)}{\cos(x) (1 + \cos(x))} \right] = \tan(x) \left[\frac{\sin^2(x)}{1 + \cos(x)} \right] = \sin^2(x) \tan(x) \left[\frac{1}{1 + \cos(x)} \right] \end{aligned}$$

This works great when we have values of x that are near 0 since then we have that $\sin(x) = \tan(x)$ and we are avoiding subtracting these two numbers.

Problem 11

$$\tan(x) - \sin(x) = \frac{\sin(x)}{\cos(x)} - \sin(x) = \sin(x) \left[\frac{1}{\cos(x)} - 1 \right]$$

So far all these are subtracting similar numbers so (a) and (c) are not good.

$$\sin(x) \left[\frac{1}{\cos(x)} - 1 \right] = \sin^2(x) \tan(x) \left[\frac{1}{\cos(x) + 1} \right]$$

This right here would be the best calculation for x values close to 0, thus we have that (f) is the best.

Since $x \approx \tan(x)$ at small x , $\frac{1}{2}x^3$ and $\frac{1}{2}x^2 \tan(x)$ are similar and better than the the first kind.

Thus, the order would be:

(a)=(c), then (b) then (e) then (d) then (f) as the best.

Problem 17

If at most 2 bits of precision are to be lost in the computation $y = \sqrt{x^2 + 1} - 1$, what restriction must be placed on x .

Using the loss of precision theorem:

$$q = 2, x = \sqrt{x^2 + 1}, y = 1$$

$$\begin{aligned}1 - \frac{1}{\sqrt{x^2 + 1}} &= 1 - \frac{y}{x} \geq 2^{-p} = 2^{-2} \\ \Rightarrow -\frac{1}{\sqrt{x^2 + 1}} &\geq \frac{1}{4} - 1 = -\frac{3}{4} \Rightarrow \frac{1}{\sqrt{x^2 + 1}} \leq \frac{3}{4} \Rightarrow \sqrt{x^2 + 1} \geq \frac{4}{3} \\ \Rightarrow x^2 + 1 &\geq \frac{16}{9} \Rightarrow x^2 \geq \frac{7}{9} \Rightarrow |x| \geq \frac{\sqrt{7}}{3}\end{aligned}$$

Computer Problems:

Problem 1

Which of the following two functions are reliable and which are not for values:

$$8^{-1}, 8^{-2}, \dots$$

Function $f(x) = \sqrt{x^2 + 1} - 1$

```
function f = f(x)
%This Program will take an array of numbers X and output
%the values of f(xi) for each xi in the array X.

%This loop will go through all the values in the array X and evaluate
%them to give our f(xi).
i = 1;
while (i < length(x) + 1)
(sqrt(x(i)*x(i)+1)-1) %This is the function f(x)
i = i + 1;

end
```

Function $g(x) = \frac{x^2}{\sqrt{x^2+1}+1}$

```
function g = g(x)
%This Program will take an array of numbers X and output
%the values of g(xj) for each xj in the array X.

%This loop will go through all the values in the array X and evaluate
%them to give our g(xj).
j= 1
while(j < length(x))
(x(j)*x(j) / (sqrt(x(j)*x(j)+1)+1)) %This is the function g(x)
j = j +1;

end
```

Running the program with the array

```
X=[8^-1, 8^-2, 8^-3, 8^-4, 8^-5, 8^-6, 8^-7, 8^-8, 8^-9, 8^-10];
```

We get that the results are:

$$f(x)$$

```
>>f(X)
ans = 0.0078
ans = 1.2206e-04
ans = 1.9073e-06
ans = 2.9802e-08
ans = 4.6566e-10
ans = 7.2760e-12
ans = 1.1369e-13
ans = 1.7764e-15
ans = 0
ans = 0
```

$$g(x)$$

```
>> g(X)
ans = 0.0078
ans = 1.2206e-04
ans = 1.9073e-06
ans = 2.9802e-08
ans = 4.6566e-10
ans = 7.2760e-12
ans = 1.1369e-13
ans = 1.7764e-15
ans = 2.7756e-17
ans = 4.3368e-19
```

We see that $g(x)$ was more accurate than $f(x)$ since we avoided subtracting like numbers. Thus $g(x)$ is more reliable for numbers closer to 0.

Problem 5

Write and test a function subprogram for $f(x) = x^{-2}(1 - \cos(x))$. Avoid loss of significance in subtraction for all arguments x and (of course) take care of the difficulty at $x = 0$.

$$f(x) = \frac{1 - \cos(x)}{x^2} = 2x^{-2} \left(\frac{1 - \cos(x)}{2} \right) = 2x^{-2} \left(\sin^2\left(\frac{x}{2}\right) \right)$$

Via small angle approximation, we know that when $|x| < 1$, we have that $\cos(x) \approx 1$ thus for these values, we will use the second formula.

Note that at $x = 0$, we are dividing by 0 thus we cannot define $f(x)$ at this point.

Here is the function:

```
function f5 = f_5(x)

%For values greater than 1, we do not have to worry about subtracting
%like numbers.
if abs(x)>1
    (1-cos(x))/(x*x)
end

%For values less than or equal to 1, we have to use the modified equation
%since we need to avoid subtracting like numbers.
if abs(x)<=1 & x ~= 0
    (2*sin(x/2)*sin(x/2))/(x*x)
end

%Since it is undefined at x=0.
if x == 0
    'UND'
end
```

Some sample outputs are:

```
>> format long
>> f_5(3)
ans = 0.221110277400049
>> f_5(.0056)
ans = 0.499998693334699
>> f_5(0)
ans = 'UND'
>> f_5(-.026)
ans = 0.499971833968015
```

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```
>> f_5(-7)
```

```
ans = 0.005022402972586
```